# Least greatest proofs

For a set of numbers $X$, how do you formalize "there is a greatest $X$" or "there is a least $X$"?

**Prove** or **disprove**: There is a least prime number.

**Prove** or **disprove**: There is a greatest integer.

*Approach 1, De Morgan's and universal generalization*:

*Approach 2, proof by contradiction*:

*Extra examples*: Prove or disprove that $\mathbb{N}$, $\mathbb{Q}$ each have a least and a greatest element.

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